LIPAS appendix numerical tables

Windows A/B/C daily hybrid vs replica RMSE context
Generated from logs/overnight_improve/final_v3
Blend: solar 0.4, radiation 0.0; storm-aware adaptive gating
Train-2 LSTM weights locked.

A: 17 Jun–14 Jul 2026 (84 site-days)
B: 18 Jul–14 Aug 2026 (84 site-days)
C: 4–17 May 2024 (42 site-days)

Window A (17 Jun – 14 Jul 2026) — hybrid daily values (page 1)

date	site	hy_sola	hy_rad	hy_temp	hy_quakes	hy_meteor	hy_dust
2026-06-17	Apollo 11 (Mare Tranquilli	10.1	0.0497	-168	43.8	2.19	1.54
2026-06-17	Artemis III candidate - No	10.4	0.0602	-162	40.8	2.11	1.49
2026-06-17	Equatorial reference (0°N	10.3	0.0496	-136	43.8	2.19	1.55
2026-06-18	Apollo 11 (Mare Tranquilli	10.1	0.0497	-170	45.8	2.19	1.54
2026-06-18	Artemis III candidate - No	9.94	0.0602	-156	42.8	2.11	1.49
2026-06-18	Equatorial reference (0°N	10	0.0496	-144	45.1	2.19	1.55
2026-06-19	Apollo 11 (Mare Tranquilli	10.1	0.0496	-172	47.8	2.19	1.57
2026-06-19	Artemis III candidate - No	10.1	0.0602	-157	44.9	2.11	1.52
2026-06-19	Equatorial reference (0°N	9.82	0.0495	-151	46.5	2.19	1.55
2026-06-20	Apollo 11 (Mare Tranquilli	10.3	0.049	-174	49.9	2.19	1.6
2026-06-20	Artemis III candidate - No	10.3	0.0598	-159	46.9	2.11	1.55
2026-06-20	Equatorial reference (0°N	10.3	0.0489	-156	48.5	2.19	1.55
2026-06-21	Apollo 11 (Mare Tranquilli	10.1	0.049	-175	51.9	2.19	1.63
2026-06-21	Artemis III candidate - No	10.5	0.0598	-163	48.9	2.11	1.58
2026-06-21	Equatorial reference (0°N	10.5	0.0489	-161	50.6	2.19	1.58
2026-06-22	Apollo 11 (Mare Tranquilli	10.9	0.0494	-151	53.5	2.19	1.66
2026-06-22	Artemis III candidate - No	10.3	0.06	-165	50.5	2.11	1.61
2026-06-22	Equatorial reference (0°N	10.4	0.0492	-166	52.3	2.19	1.61
2026-06-23	Apollo 11 (Mare Tranquilli	10.2	0.0492	-11.2	51.4	2.19	1.62
2026-06-23	Artemis III candidate - No	9.82	0.0599	-161	48.5	2.11	1.58
2026-06-23	Equatorial reference (0°N	10.1	0.0491	-170	51.7	2.19	1.64
2026-06-24	Apollo 11 (Mare Tranquilli	10.9	0.0522	35.4	49.4	2.19	1.59
2026-06-24	Artemis III candidate - No	10.8	0.0618	-156	46.4	2.11	1.54
2026-06-24	Equatorial reference (0°N	10.4	0.0521	-99.1	50.8	2.19	1.66
2026-06-25	Apollo 11 (Mare Tranquilli	10.1	0.057	67.3	47.4	2.19	1.56
2026-06-25	Artemis III candidate - No	10.1	0.0649	-155	44.4	2.11	1.51
2026-06-25	Equatorial reference (0°N	10.1	0.0569	1.56	48.8	2.19	1.63
2026-06-26	Apollo 11 (Mare Tranquilli	9.73	0.0583	91.9	45.3	2.19	1.54
2026-06-26	Artemis III candidate - No	9.73	0.0658	-145	42.4	2.11	1.49
2026-06-26	Equatorial reference (0°N	9.73	0.0582	44.8	46.8	2.19	1.6
2026-06-27	Apollo 11 (Mare Tranquilli	10.2	0.0552	107	43.7	2.19	1.54
2026-06-27	Artemis III candidate - No	10.2	0.0638	-119	40.7	2.11	1.49
2026-06-27	Equatorial reference (0°N	10.2	0.0551	72.5	45	2.19	1.56
2026-06-28	Apollo 11 (Mare Tranquilli	9.79	0.0545	115	43.7	2.19	1.54
2026-06-28	Artemis III candidate - No	9.79	0.0634	-107	40.7	2.11	1.49

Window A (17 Jun – 14 Jul 2026) — hybrid daily values (page 2)

date	site	hy_solar	hy_rad	hy_temp	hy_quakes	hy_meteor	hy_dust
2026-06-28	Equatorial reference (0°N	9.79	0.0544	91.5	44.3	2.19	1.55
2026-06-29	Apollo 11 (Mare Tranquilli	9.99	0.0541	120	43.7	2.19	1.54
2026-06-29	Artemis III candidate - No	9.99	0.0631	-103	40.7	2.11	1.49
2026-06-29	Equatorial reference (0°N	9.99	0.054	104	43.7	2.19	1.55
2026-06-30	Apollo 11 (Mare Tranquilli	12.2	0.0616	117	43.7	2.19	1.54
2026-06-30	Artemis III candidate - No	12.2	0.0679	-105	40.7	2.12	1.49
2026-06-30	Equatorial reference (0°N	12.2	0.0615	112	43.7	2.19	1.55
2026-07-01	Apollo 11 (Mare Tranquilli	33.6	0.128	111	43.7	2.2	1.54
2026-07-01	Artemis III candidate - No	33.6	0.137	-107	40.7	2.13	1.5
2026-07-01	Equatorial reference (0°N	33.6	0.128	115	43.7	2.2	1.55
2026-07-02	Apollo 11 (Mare Tranquilli	12.7	0.0581	99.5	44.2	2.22	1.54
2026-07-02	Artemis III candidate - No	12.7	0.0656	-104	41.3	2.14	1.5
2026-07-02	Equatorial reference (0°N	12.7	0.058	114	44.1	2.22	1.55
2026-07-03	Apollo 11 (Mare Tranquilli	14.3	0.0599	83	46.3	2.24	1.55
2026-07-03	Artemis III candidate - No	14.3	0.0668	-102	43.3	2.16	1.5
2026-07-03	Equatorial reference (0°N	14.3	0.0598	108	45.4	2.24	1.55
2026-07-04	Apollo 11 (Mare Tranquilli	42.6	0.178	54.1	48.3	2.26	1.58
2026-07-04	Artemis III candidate - No	42.6	0.19	-104	45.4	2.19	1.53
2026-07-04	Equatorial reference (0°N	42.6	0.178	97.1	47	2.26	1.55
2026-07-05	Apollo 11 (Mare Tranquilli	25.4	0.101	21.2	50.3	2.29	1.62
2026-07-05	Artemis III candidate - No	25.4	0.109	-107	47.4	2.21	1.57
2026-07-05	Equatorial reference (0°N	25.4	0.101	80.7	49	2.29	1.56
2026-07-06	Apollo 11 (Mare Tranquilli	10.1	0.0502	-42.9	52.4	2.32	1.65
2026-07-06	Artemis III candidate - No	10.3	0.0606	-113	49.4	2.24	1.6
2026-07-06	Equatorial reference (0°N	10.1	0.0501	56.7	51.1	2.32	1.6
2026-07-07	Apollo 11 (Mare Tranquilli	9.24	0.052	-95.4	53	2.33	1.66
2026-07-07	Artemis III candidate - No	10.2	0.0617	-105	50	2.25	1.61
2026-07-07	Equatorial reference (0°N	10.1	0.0519	20.9	52.2	2.33	1.63
2026-07-08	Apollo 11 (Mare Tranquilli	9.86	0.0537	-129	51	2.34	1.63
2026-07-08	Artemis III candidate - No	10	0.0628	-111	48	2.26	1.58
2026-07-08	Equatorial reference (0°N	9.64	0.0536	-43.6	51.6	2.34	1.66
2026-07-09	Apollo 11 (Mare Tranquilli	9.77	0.0548	-136	48.9	2.33	1.6
2026-07-09	Artemis III candidate - No	9.86	0.0636	-124	46	2.25	1.55
2026-07-09	Equatorial reference (0°N	9.55	0.0548	-29.8	50.3	2.33	1.67
2026-07-10	Apollo 11 (Mare Tranquilli	10.2	0.0542	-143	46.9	2.31	1.56

Window A (17 Jun – 14 Jul 2026) — hybrid daily values (page 3)

date	site	hy_solar	hy_rad	hy_temp	hy_quakes	hy_meteor	hy_dust
2026-07-10	Artemis III candidate - No	9.82	0.0632	-133	43.9	2.23	1.52
2026-07-10	Equatorial reference (0°N	10.2	0.0541	-50.4	48.3	2.31	1.63
2026-07-11	Apollo 11 (Mare Tranquilli	6.76	0.056	-149	44.9	2.28	1.55
2026-07-11	Artemis III candidate - No	6.76	0.0643	-134	41.9	2.2	1.5
2026-07-11	Equatorial reference (0°N	6.76	0.056	-68.7	46.3	2.28	1.6
2026-07-12	Apollo 11 (Mare Tranquilli	10.2	0.0541	-153	43.7	2.24	1.55
2026-07-12	Artemis III candidate - No	10.5	0.0631	-139	40.7	2.17	1.5
2026-07-12	Equatorial reference (0°N	10.2	0.054	-84.9	44.8	2.24	1.56
2026-07-13	Apollo 11 (Mare Tranquilli	10.1	0.0544	-158	43.7	2.21	1.54
2026-07-13	Artemis III candidate - No	9.63	0.0633	-152	40.7	2.14	1.5
2026-07-13	Equatorial reference (0°N	10.2	0.0543	-99.2	44.1	2.21	1.55
2026-07-14	Apollo 11 (Mare Tranquilli	10.2	0.0544	-161	43.7	2.19	1.54
2026-07-14	Artemis III candidate - No	10.2	0.0633	-160	40.7	2.11	1.49
2026-07-14	Equatorial reference (0°N	9.84	0.0543	-112	43.7	2.19	1.55

Window B (18 Jul – 14 Aug 2026) — hybrid daily values (page 1)

date	site	hy_solar	hy_rad	hy_temp	hy_quakes	hy_meteor	hy_dust
2026-07-18	Apollo 11 (Mare Tranquilli	10.1	0.145	-171	46.8	2.13	1.54
2026-07-18	Artemis III candidate - No	9.87	0.155	-156	43.8	2.06	1.5
2026-07-18	Equatorial reference (0°N	10.5	0.145	-147	45.7	2.13	1.54
2026-07-19	Apollo 11 (Mare Tranquilli	11.5	0.0484	-173	47.9	2.13	1.58
2026-07-19	Artemis III candidate - No	12.5	0.0594	-160	47	2.06	1.55
2026-07-19	Equatorial reference (0°N	11.7	0.0483	-153	46.6	2.13	1.54
2026-07-20	Apollo 11 (Mare Tranquilli	12.8	0.0513	-175	49.1	2.12	1.61
2026-07-20	Artemis III candidate - No	12.6	0.0613	-163	48.1	2.05	1.58
2026-07-20	Equatorial reference (0°N	12.6	0.0512	-159	47.8	2.12	1.55
2026-07-21	Apollo 11 (Mare Tranquilli	13.4	0.057	-176	50.3	2.12	1.64
2026-07-21	Artemis III candidate - No	13.5	0.065	-164	48.8	2.05	1.6
2026-07-21	Equatorial reference (0°N	13	0.0569	-163	49	2.12	1.58
2026-07-22	Apollo 11 (Mare Tranquilli	13.4	0.055	-79.7	50.1	2.11	1.63
2026-07-22	Artemis III candidate - No	13.4	0.0637	-161	47.6	2.04	1.57
2026-07-22	Equatorial reference (0°N	13.7	0.0549	-168	49.6	2.11	1.62
2026-07-23	Apollo 11 (Mare Tranquilli	12.9	0.055	16.6	48.9	2.11	1.6
2026-07-23	Artemis III candidate - No	12.8	0.0637	-155	46.5	2.04	1.53
2026-07-23	Equatorial reference (0°N	13.7	0.0549	-172	49.8	2.11	1.65
2026-07-24	Apollo 11 (Mare Tranquilli	13.5	0.055	51	47.7	2.11	1.57
2026-07-24	Artemis III candidate - No	13	0.0637	-155	45.3	2.04	1.5
2026-07-24	Equatorial reference (0°N	13	0.0549	-35.6	49.1	2.11	1.64
2026-07-25	Apollo 11 (Mare Tranquilli	13.3	0.055	81.3	46.6	2.11	1.54
2026-07-25	Artemis III candidate - No	13.7	0.0637	-142	44.1	2.04	1.49
2026-07-25	Equatorial reference (0°N	13.6	0.0549	24.7	48	2.11	1.61
2026-07-26	Apollo 11 (Mare Tranquilli	13	0.055	99.9	45.4	2.12	1.53
2026-07-26	Artemis III candidate - No	13.4	0.0637	-118	43.3	2.05	1.49
2026-07-26	Equatorial reference (0°N	13.3	0.0549	59.3	46.8	2.12	1.57
2026-07-27	Apollo 11 (Mare Tranquilli	13.6	0.055	111	45	2.14	1.54
2026-07-27	Artemis III candidate - No	12.9	0.0637	-106	43.3	2.06	1.49
2026-07-27	Equatorial reference (0°N	13	0.0549	82.5	45.9	2.14	1.54
2026-07-28	Apollo 11 (Mare Tranquilli	13.2	0.054	117	45	2.17	1.54
2026-07-28	Artemis III candidate - No	13.3	0.063	-103	43.3	2.1	1.49
2026-07-28	Equatorial reference (0°N	13.3	0.0539	98.4	45.2	2.17	1.55
2026-07-29	Apollo 11 (Mare Tranquilli	13.1	0.0535	120	45	2.22	1.54
2026-07-29	Artemis III candidate - No	13.7	0.0627	-106	43.3	2.15	1.5

Window B (18 Jul – 14 Aug 2026) — hybrid daily values (page 2)

date	site	hy_solar	hy_rad	hy_temp	hy_quakes	hy_meteor	hy_dust
2026-07-29	Equatorial reference (0°N	13.1	0.0534	109	45	2.22	1.55
2026-07-30	Apollo 11 (Mare Tranquilli	13.5	0.0535	115	45	2.3	1.55
2026-07-30	Artemis III candidate - No	13.8	0.0627	-106	43.3	2.22	1.5
2026-07-30	Equatorial reference (0°N	13.1	0.0534	114	45	2.3	1.56
2026-07-31	Apollo 11 (Mare Tranquilli	13.4	0.0535	106	45	2.39	1.56
2026-07-31	Artemis III candidate - No	12.9	0.0627	-104	43.7	2.31	1.51
2026-07-31	Equatorial reference (0°N	13.1	0.0534	115	45	2.39	1.57
2026-08-01	Apollo 11 (Mare Tranquilli	13.7	0.0535	92.2	45.9	2.49	1.57
2026-08-01	Artemis III candidate - No	13.9	0.0627	-102	44.9	2.41	1.53
2026-08-01	Equatorial reference (0°N	13.4	0.0534	112	45.4	2.49	1.58
2026-08-02	Apollo 11 (Mare Tranquilli	12.9	0.0553	69.8	47	2.58	1.6
2026-08-02	Artemis III candidate - No	13.1	0.0639	-104	46.1	2.49	1.57
2026-08-02	Equatorial reference (0°N	13.2	0.0553	104	45.8	2.58	1.59
2026-08-03	Apollo 11 (Mare Tranquilli	13.6	0.0591	39.3	48.2	2.64	1.63
2026-08-03	Artemis III candidate - No	13.8	0.0663	-108	47.2	2.55	1.61
2026-08-03	Equatorial reference (0°N	13.6	0.059	90.5	46.9	2.64	1.59
2026-08-04	Apollo 11 (Mare Tranquilli	12.8	0.0535	1.87	49.4	2.66	1.67
2026-08-04	Artemis III candidate - No	13.2	0.0627	-114	48.4	2.57	1.64
2026-08-04	Equatorial reference (0°N	13	0.0534	70.9	48.1	2.66	1.61

Window C (4 – 17 May 2024) — hybrid daily values (page 1)

date	site	hy_solar	hy_rad	hy_temp	hy_quakes	hy_meteor	hy_dust
2024-05-04	Apollo 11 (Mare Tranquilli	15	0.0599	-145	46.2	2.48	1.57
2024-05-04	Artemis III candidate - No	15	0.0668	-135	43.3	2.4	1.52
2024-05-04	Equatorial reference (0°N	15	0.0599	-56.6	47.6	2.48	1.64
2024-05-05	Apollo 11 (Mare Tranquilli	26.4	0.0994	-150	44.2	2.47	1.57
2024-05-05	Artemis III candidate - No	26.4	0.107	-134	41.2	2.39	1.52
2024-05-05	Equatorial reference (0°N	26.4	0.0994	-74.2	45.6	2.47	1.61
2024-05-06	Apollo 11 (Mare Tranquilli	17	0.0745	-155	43.7	2.43	1.57
2024-05-06	Artemis III candidate - No	17	0.0799	-144	40.7	2.35	1.52
2024-05-06	Equatorial reference (0°N	17	0.0745	-89.8	44.6	2.43	1.57
2024-05-07	Apollo 11 (Mare Tranquilli	28.2	0.105	-159	43.7	2.37	1.56
2024-05-07	Artemis III candidate - No	28.2	0.113	-154	40.7	2.29	1.51
2024-05-07	Equatorial reference (0°N	28.2	0.105	-103	43.9	2.37	1.56
2024-05-08	Apollo 11 (Mare Tranquilli	38.9	0.139	-162	43.7	2.29	1.55
2024-05-08	Artemis III candidate - No	38.9	0.149	-158	40.7	2.22	1.5
2024-05-08	Equatorial reference (0°N	38.9	0.139	-115	43.7	2.3	1.56
2024-05-09	Apollo 11 (Mare Tranquilli	40.5	0.144	-164	43.7	2.23	1.55
2024-05-09	Artemis III candidate - No	40.5	0.154	-152	40.7	2.15	1.5
2024-05-09	Equatorial reference (0°N	40.5	0.144	-126	43.7	2.23	1.55
2024-05-10	Apollo 11 (Mare Tranquilli	33.7	0.132	-167	43.7	2.18	1.54
2024-05-10	Artemis III candidate - No	33.7	0.141	-161	40.7	2.1	1.49
2024-05-10	Equatorial reference (0°N	33.7	0.132	-135	43.7	2.18	1.55
2024-05-11	Apollo 11 (Mare Tranquilli	37.8	0.245	-170	45.4	2.14	1.54
2024-05-11	Artemis III candidate - No	37.8	0.263	-157	42.4	2.07	1.49
2024-05-11	Equatorial reference (0°N	37.8	0.245	-143	44.8	2.14	1.54
2024-05-12	Apollo 11 (Mare Tranquilli	27.2	0.113	-172	47.4	2.12	1.55
2024-05-12	Artemis III candidate - No	27.2	0.121	-156	44.5	2.04	1.51
2024-05-12	Equatorial reference (0°N	27.2	0.113	-149	46.2	2.12	1.54
2024-05-13	Apollo 11 (Mare Tranquilli	25.6	0.103	-173	49.5	2.1	1.58
2024-05-13	Artemis III candidate - No	25.6	0.11	-159	46.5	2.03	1.54
2024-05-13	Equatorial reference (0°N	25.6	0.103	-155	48.1	2.1	1.54
2024-05-14	Apollo 11 (Mare Tranquilli	19.3	0.0769	-175	51.5	2.1	1.62
2024-05-14	Artemis III candidate - No	19.3	0.0825	-162	48.5	2.03	1.57
2024-05-14	Equatorial reference (0°N	19.3	0.0769	-160	50.2	2.1	1.56
2024-05-15	Apollo 11 (Mare Tranquilli	17.5	0.0733	-177	53.5	2.1	1.65
2024-05-15	Artemis III candidate - No	17.5	0.0786	-165	50.6	2.03	1.6

Window C (4 – 17 May 2024) — hybrid daily values (page 2)

date	site	hy_solar	hy_rad	hy_temp	hy_quakes	hy_meteor	hy_dust
2024-05-15	Equatorial reference (0°N	17.5	0.0733	-165	52.2	2.1	1.59
2024-05-16	Apollo 11 (Mare Tranquilli	12.5	0.0605	-36.9	51.8	2.1	1.62
2024-05-16	Artemis III candidate - No	12.5	0.0672	-162	48.9	2.03	1.57
2024-05-16	Equatorial reference (0°N	12.5	0.0605	-169	51.8	2.1	1.63
2024-05-17	Apollo 11 (Mare Tranquilli	12.2	0.0602	28.5	49.8	2.11	1.59
2024-05-17	Artemis III candidate - No	12.2	0.067	-157	46.9	2.04	1.54
2024-05-17	Equatorial reference (0°N	12.2	0.0601	-173	51.2	2.11	1.66